

50 LINUX COMMANDS - Notes

BASIC LINUX COMMANDS

1. pwd – Print Working Directory

Full Form: Print Working Directory

Definition: Shows the current directory you are in.

Example: pwd

2. ls – List

Full Form: List Directory Contents

Definition: Displays all files and folders inside a directory.

Example: ls -la

3. cd – Change Directory

Full Form: Change Directory

Definition: Moves you from one directory to another.

Example: cd /etc

4. clear – Clear Terminal Screen

Full Form: Clear

Definition: Clears terminal output for clean workspace.

Example: clear

5. echo – Echo Text

Full Form: Echo (Repeats text)

Definition: Prints text on screen or writes text to a file.

Example: echo "Hello" > file.txt

6. date – Display Date

Full Form: Date

Definition: Shows current date and time.

Example: date

7. man – Manual

Full Form: Manual

Definition: Displays documentation/help for any command.

Example: man nmap

8. exit – Exit Terminal

Definition: Closes the terminal session or SSH session.

Example: exit

FILE & DIRECTORY MANAGEMENT COMMANDS

9. mkdir – Make Directory

Full Form: Make Directory

Definition: Creates a new folder.

Example: mkdir recon

10. rmdir – Remove Directory

Full Form: Remove Directory

Definition: Deletes an empty directory.

Example: rmdir testfolder

11. touch – Touch File

Full Form: Touch (Updates timestamp / creates file)

Definition: Creates an empty file (0 byte).

Example: touch notes.txt

12. cp – Copy

Full Form: Copy

Definition: Copies files or directories.

Example: cp file.txt backup.txt

13. mv – Move

Full Form: Move

Definition: Moves or renames files.

Example: mv file.txt /tmp/

14. rm – Remove

Full Form: Remove

Definition: Deletes files and folders permanently.

Example: rm file.txt

15. cat – Concatenate

Full Form: Concatenate

Definition: Displays file contents.

Example: cat /etc/passwd

16. head – Show First Lines

Definition: Displays first 10 lines of a file.

Example: head -n 20 auth.log

17. tail – Show Last Lines

Definition: Displays last 10 lines of a file.

Example: tail -f syslog

18. less – Read Files Slowly

Definition: Opens long files page by page.

Example: less /etc/services

19. find – Find Files

Definition: Searches for files by name, size, permissions.

Example: find / -name passwd 2>/dev/null

20. locate – Locate Files Quickly

Definition: Quickly find files using a database.

Example: locate ssh_config

PERMISSION & USER MANAGEMENT COMMANDS

21. chmod – Change Mode

Full Form: Change Mode

Definition: Changes file permissions (read, write, execute).

Example: chmod 755 script.sh

22. chown – Change Owner

Full Form: Change Owner

Definition: Changes the file's owner and group.

Example: chown root:root file.txt

23. chgrp – Change Group

Full Form: Change Group

Definition: Changes group ownership of a file.

Example: chgrp admin file.txt

24. umask – User Mask

Full Form: User Mask

Definition: Sets default permissions for new files.

Example: umask 022

25. sudo – SuperUser Do

Full Form: SuperUser Do

Definition: Runs commands with administrator/root privileges.

Example: sudo apt update

26. su – Switch User

Full Form: Switch User

Definition: Switches to another user or root.

Example: su root

27. passwd – Password Change

Definition: Changes user password.

Example: passwd

INFORMATION & PROCESS MANAGEMENT

28. whoami – Who Am I

Definition: Shows currently logged-in user.

Example: whoami

29. id – Identity

Definition: Shows user ID, group ID, and permissions.

Example: id

30. uname – Unix Name

Definition: Shows system and kernel information.

Example: uname -a

31. hostname – Host Name

Definition: Displays machine hostname.

Example: hostname

32. df – Disk Free

Full Form: Disk Free

Definition: Shows available disk space.

Example: df -h

33. du – Disk Usage

Full Form: Disk Usage

Definition: Displays size of files and directories.

Example: du -sh /var/log

34. ps – Process Status

Full Form: Process Status

Definition: Shows currently running processes.

Example: ps aux

35. top – Table Of Processes

Full Form: Table Of Processes

Definition: Displays real-time system resource usage.

Example: top

36. kill – Stop Process

Definition: Terminates a process using PID.

Example: kill 1050

37. history – Command History

Definition: Shows last executed commands.

Example: history

NETWORKING + CYBERSECURITY COMMANDS

38. ip – Internet Protocol

Full Form: Internet Protocol Utility

Definition: Shows network interfaces & IP address.

Example: ip a

39. ifconfig – Interface Configuration

Full Form: Interface Configuration

Definition: Displays network interfaces (older tool).

Example: ifconfig

40. ping – Packet Internet Groper

Full Form: Packet Internet Groper

Definition: Tests connectivity between two devices.

Example: ping google.com

41. netstat – Network Statistics

Full Form: Network Statistics

Definition: Shows open ports and network connections.

Example: netstat -tulnp

42. ss – Socket Statistics

Full Form: Socket Statistics

Definition: Shows detailed socket connections (faster than netstat).

Example: ss -tulnp

43. arp – Address Resolution Protocol

Definition: Displays ARP table (local network devices).

Example: arp -a

44. traceroute – Trace Route

Full Form: Trace Route

Definition: Shows path packets take to reach destination.

Example: traceroute google.com

45. curl – Client URL

Full Form: Client URL

Definition: Used to send HTTP requests or download data.

Example: curl http://example.com

46. wget – Web Get

Full Form: Web Get

Definition: Downloads files from internet.

Example: wget http://site.com/file.txt

47. ssh – Secure Shell

Full Form: Secure Shell

Definition: Connects securely to remote systems.

Example: ssh user@192.168.1.10

48. scp – Secure Copy Protocol

Full Form: Secure Copy

Definition: Transfers files securely using SSH.

Example: scp file.txt user@host:/tmp/

49. nmap – Network Mapper

Full Form: Network Mapper

Definition: Performs port scanning, service detection, OS detection.

Example: nmap -sV 10.10.10.10

50. dig – Domain Information Groper

Full Form: Domain Information Groper

Definition: Used for DNS lookup, domain info.

Example: dig google.com

PACKAGE MANAGEMENT COMMANDS

51. apt – Advanced Package Tool

Full Form: Advanced Package Tool

Definition: Manages software installation and updates (Debian/Kali/Ubuntu).

Example: `sudo apt install nmap`

52. apt update – Refresh Package List

Definition: Updates list of available packages.

Example: `sudo apt update`

53. apt upgrade – Upgrade Installed Packages

Definition: Updates all installed tools.

Example: `sudo apt upgrade`

54. apt remove – Uninstall Package

Definition: Removes installed software.

Example: `sudo apt remove apache2`

55. dpkg – Debian Package

Full Form: Debian Package

Definition: Installs .deb files manually.

Example: `sudo dpkg -i tool.deb`

56. snap – Snappy Package Manager

Definition: Installs snap-based software.

Example: `sudo snap install code`

57. apt autoremove – Cleanup Unused Packages

Definition: Removes leftover dependencies.

Example: `sudo apt autoremove`

